DATABASE MANAGEMENT SYSTEMS • PROJECT-BASED LEARNING

PROJECT 31

Event Registration & Venue Scheduling System

Design and Implementation of a Relational Database Management System for institutional event, venue and session administration

Scope • ER Diagram • Initial Schema

What This Review Covers

01

Problem Statement

Why colleges need this system

02

Scope & Objectives

What Review 1 sets out to deliver

03

Users & Roles

Who interacts with the system

04

Functional Requirements

Core capabilities in scope

05

Key Entities

The 12 building blocks of the ER model

06

ER Diagram

How the entities relate

07

Initial Schema

Proposed tables, keys & business rules

01 Problem Statement

Institutions regularly organize seminars, workshops and cultural events — each requiring coordinated tracking of venues, sessions, speakers, participants, payments, attendance and certificates.

When handled manually or through disconnected spreadsheets, this produces real operational failures:

- **Venue double-booking**
  Two sessions scheduled in the same room at the same time
- **Duplicate registrations**
  Participants signing up more than once for an event
- **Unverifiable attendance**
  No reliable basis for issuing certificates

WHY A DBMS

A single normalized database — with enforced keys and constraints — replaces scattered records with one consistent source of truth for every event, venue, session and participant.

02 Scope & Objectives

PROJECT SCOPE

- Covers institutional seminars, workshops and cultural events end-to-end
- Spans event & venue setup through registration, attendance and certification
- Targets a MySQL / PostgreSQL / Oracle backend with a Java or Python front-end

OBJECTIVES

1. Identify entities, attributes, relationships and user roles
2. Design a conceptual ER model and normalize it to 3NF
3. Enforce integrity via primary/foreign keys and CHECK constraints
4. Build a working CRUD application with meaningful reports

03 Users & Roles

EC

Event Coordinator

Sets up events, sessions and allocates venues; owns event-level scheduling

SP

Speaker

Assigned to one or more sessions; needs schedule visibility

PT

Participant

Registers for events, pays fees, attends sessions, earns certificates

AD

Admin / Finance

Oversees payments, monitors attendance and generates reports

04 Functional Requirements

1

Event & Session Setup

Create events, categories and sessions; assign coordinators and speakers

2

Venue Allocation

Assign venues to sessions with capacity and overlap checks

3

Participant Registration

Register participants against events with duplicate prevention

4

Payment Tracking

Record registration payments; enforce non-negative amounts

5

Attendance Logging

Capture per-session attendance for eligibility checks

6

Certificate & Reporting

Issue certificates on threshold and generate feedback/revenue reports

05 Conceptual Design — Key Entities

The conceptual ER model represents the following 12 entities, to be normalized to 3NF in Review 2.

1 Events

Master event records

2 Categories

Event classification

3 Venues

Physical locations & capacity

4 Sessions

Time-slots within an event

5 Coordinators

Staff managing an event

6 Speakers

Assigned to sessions

7 Participants

Registered attendees

8 Registrations

Participant ↔ event link

9 Payments

Registration fee records

10 Attendance

Session check-in log

11 Certificates

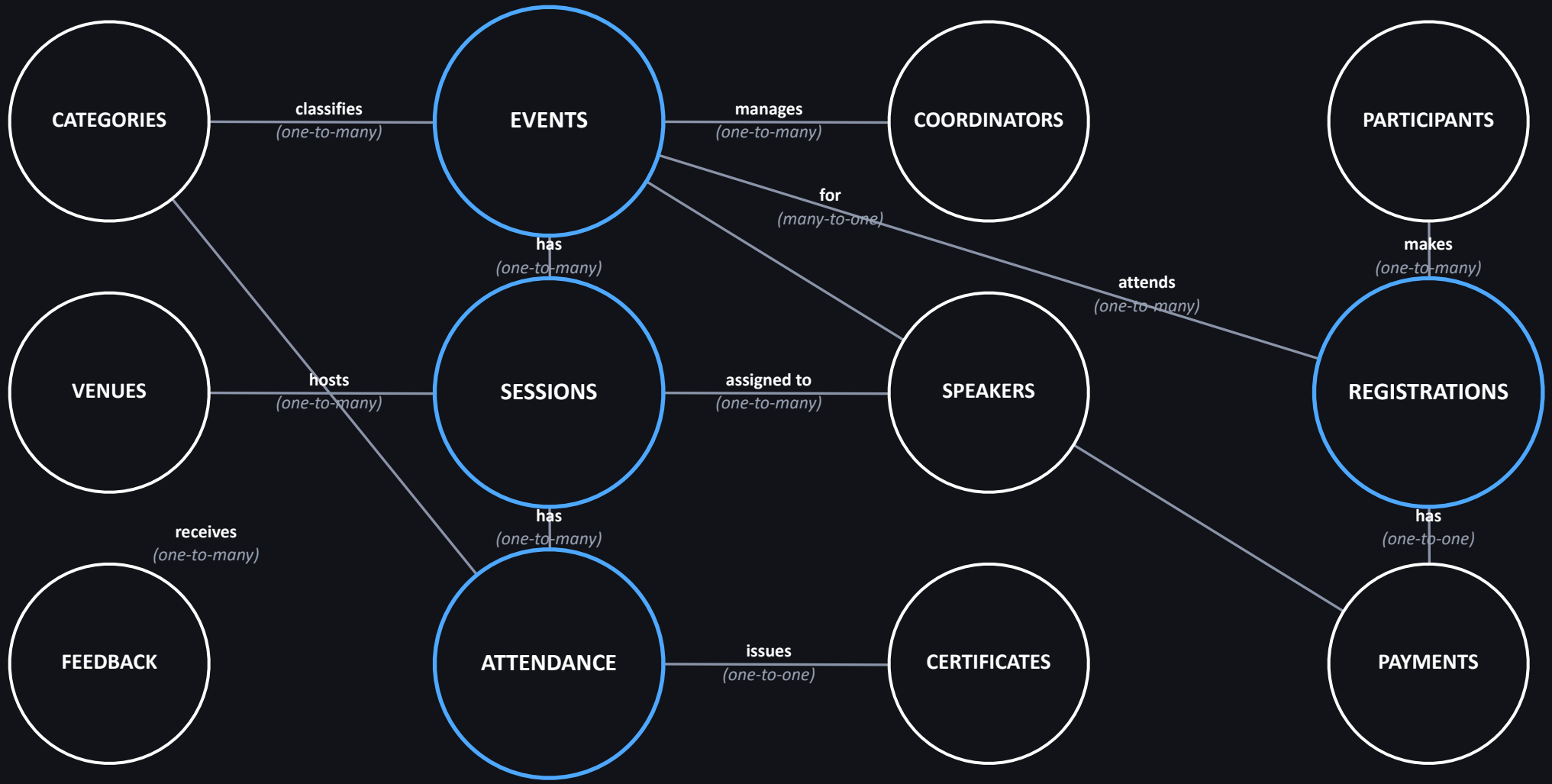
Issued on eligibility

12 Feedback

Post-event responses

Conceptual Design — ER Diagram

Principal Entities and Relationships (Crow's Foot Notation)



Entities: CATEGORIES, EVENTS, COORDINATORS, PARTICIPANTS, VENUES, SESSIONS, SPEAKERS, REGISTRATIONS, FEEDBACK, ATTENDANCE, CERTIFICATES, PAYMENTS

07 Initial Relational Schema

PROPOSED TABLES & PRIMARY KEYS

- **Events / Categories / Venues**

  event_id, category_id, venue_id (PK)

- **Sessions**

  session_id (PK); FK → event_id, venue_id, speaker_id

- **Coordinators / Speakers / Participants**

  coordinator_id, speaker_id, participant_id (PK)

- **Registrations**

  registration_id (PK); FK → participant_id, event_id

- **Payments**

  payment_id (PK); FK → registration_id

- **Attendance / Certificates / Feedback**

  attendance_id, certificate_id, feedback_id (PK)

KEY BUSINESS RULES TO ENFORCE

- **No venue overlap**

  A venue can't host two overlapping sessions

- **Capacity limits**

  Registrations can't exceed venue/session capacity

- **No duplicate registration**

  One participant, one registration per event

- **Attendance threshold**

  Certificates issued only above minimum attendance

- **Non-negative payments**

  Payment amounts must be ≥ 0

08 Deliverables

1. Problem identification, scope, objectives, users and functional requirements

2. Conceptual ER diagram covering all 12 principal entities and their relationships

3. Initial relational schema: proposed tables, primary keys and relationships